

First Boot

```
Press <ENTER> to continue.
VirtualBox Virtual Machine - Netgate Device ID: e64bdec8d502b5cc6

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

WAN (wan) -> em0 -> v4/DHCP4: 10.0.2.15/24
                v6/DHCP6: fd17:625c:f037:2:a00:27ff:fee3:a067/64
LAN (lan) -> em1 -> v4: 192.168.56.2/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                    14) Enable Secure Shell (sshd)
6) Halt system                      15) Restore recent configuration
7) Ping host                        16) Restart PHP-FPM
8) Shell

Enter an option: ^[
Message from syslogd@pfSense at Jun 22 15:52:52 ...
php-fpm[4331]: /index.php: Successful login for user 'admin' from: 192.168.56.1 (
Local Database)
^R
```

Configure LAN IP Address

```
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via DHCP6? (y/n) n

Enter the new LAN IPv6 address. Press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? (y/n) n
Disabling IPv4 DHCPD...
Disabling IPv6 DHCPD...

Do you want to revert to HTTP as the webConfigurator protocol? (y/n) n

Please wait while the changes are saved to LAN...
Reloading filter...
Reloading routing configuration...
DHCPD...

The IPv4 LAN address has been set to 192.168.25.5/24
You can now access the webConfigurator by opening the following URL in your web
browser:
                https://192.168.25.5/

Press <ENTER> to continue. █
```

The IPv4 LAN address has been set to 192.168.25.5/24
You can now access the webConfigurator by opening the following URL in your web browser:

`https://192.168.25.5/`

Press <ENTER> to continue.

VMware Virtual Machine - Netgate Device ID: 24273dea170cfb56a452

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

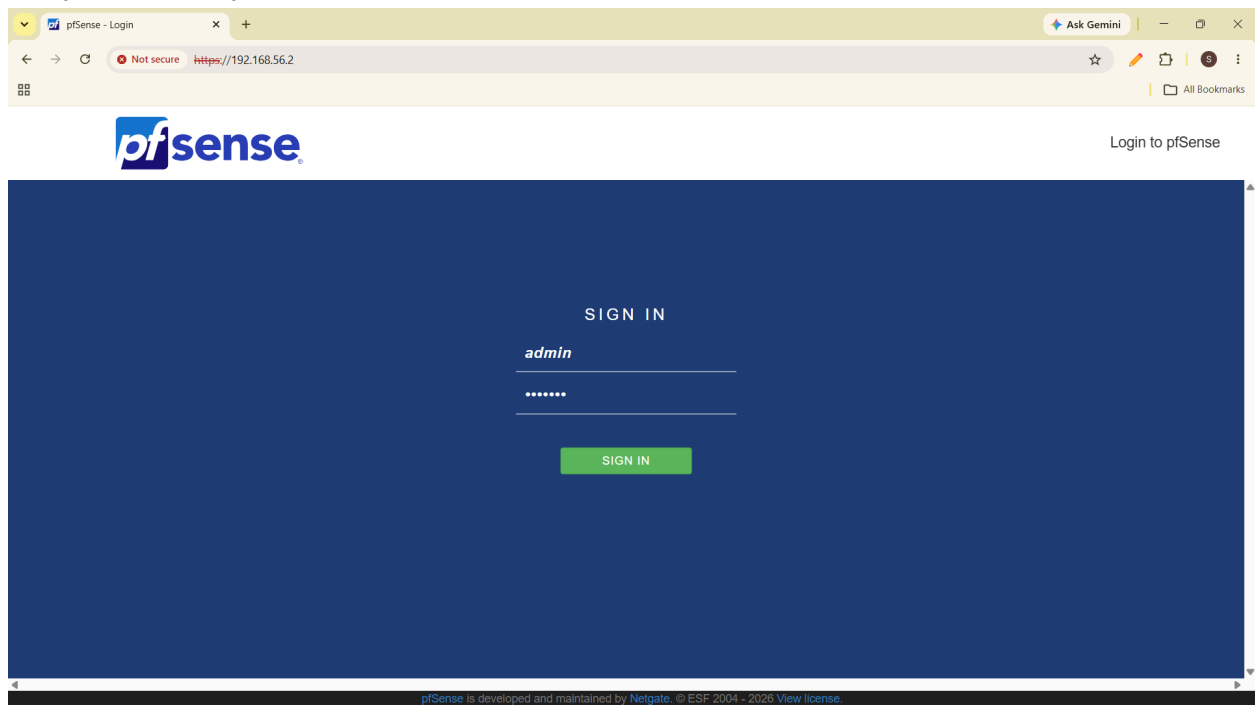
WAN (wan) -> em0 -> v4/DHCP4: 192.168.74.238/24

LAN (lan) -> em1 -> v4: 192.168.25.5/24

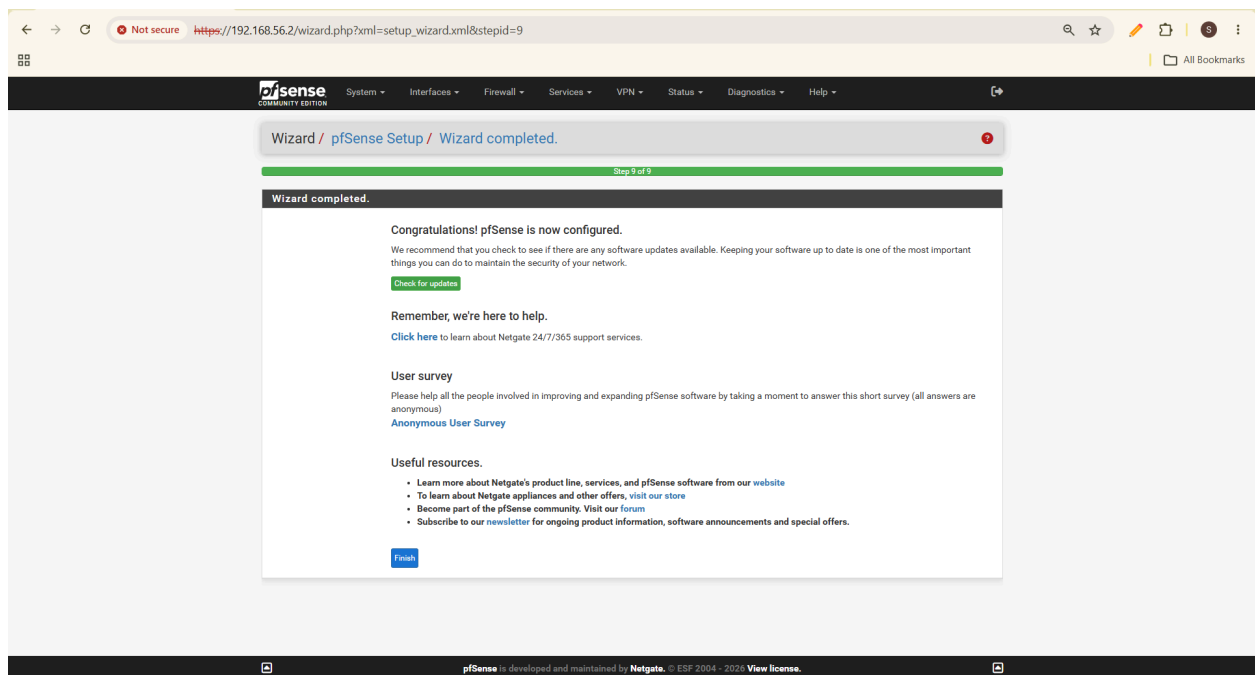
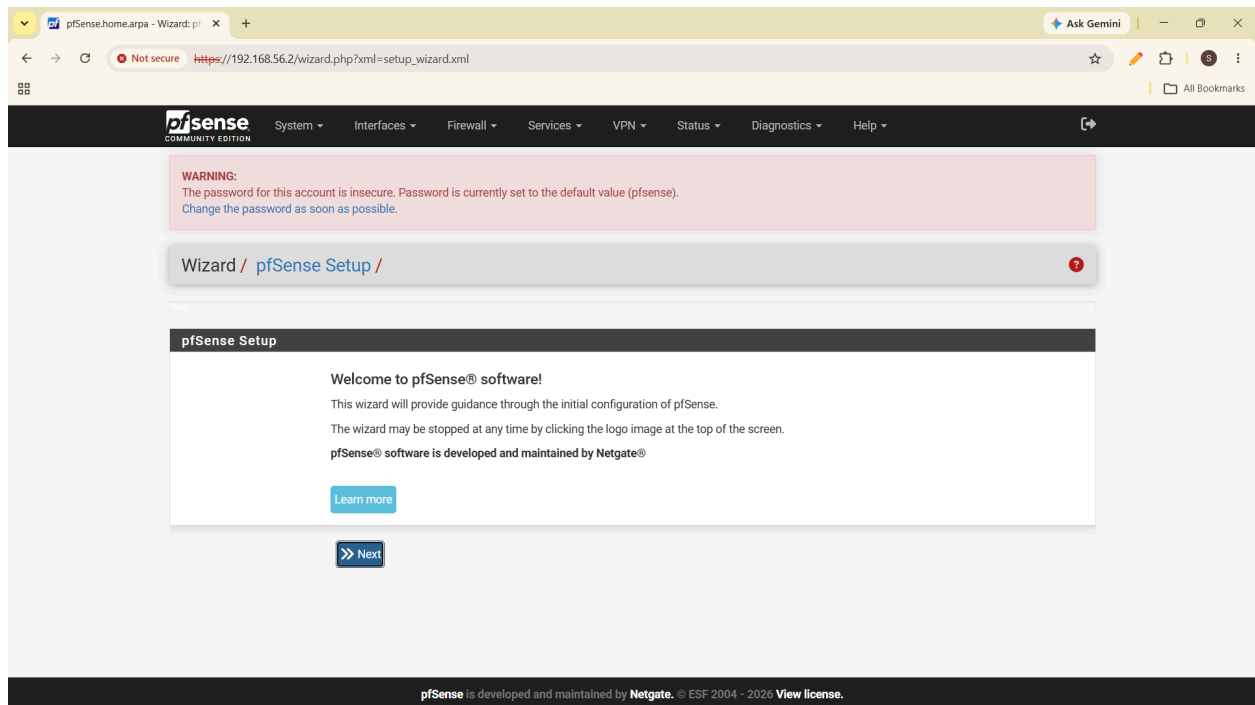
- | | |
|-------------------------------------|----------------------------------|
| 0) Logout / Disconnect SSH | 9) pfTop |
| 1) Assign Interfaces | 10) Filter Logs |
| 2) Set interface(s) IP address | 11) Restart GUI |
| 3) Reset admin account and password | 12) PHP shell + pfSense tools |
| 4) Reset to factory defaults | 13) Update from console |
| 5) Reboot system | 14) Enable Secure Shell (sshd) |
| 6) Halt system | 15) Restore recent configuration |
| 7) Ping host | 16) Restart PHP-FPM |
| 8) Shell | |

Enter an option: █

Verify Connectivity and access web interface



Login



← → ↻ Not secure https://192.168.56.2

pfSense
community edition

System Interfaces Firewall Services VPN Status Diagnostics Help

Status / Dashboard

System Information

Name	pfSense.home.apa
User	admin@192.168.56.1 (Local Database)
System	VirtualBox Virtual Machine Netgate Device ID: e64bdecdac8d502b5cc6
BIOS	Vendor: innotek GmbH Version: VirtualBox Release Date: Fri Dec 1 2006 Boot Method: BIOS
Version	2.8.1-RELEASE (amd64) built on Thu May 14 11:13:00 IST 2026 FreeBSD 15.0-CURRENT The system is on the latest version. Version information updated at Mon Jun 22 21:09:02 IST 2026
CPU Type	Intel(R) Core(TM) i7-8650U CPU @ 1.90GHz 2 CPUs x 1 package(s) x 2 cache groups x 1 core(s) AES-NI CPU Crypto: Yes (inactive) QAT Crypto: No
Hardware crypto	Inactive
Kernel PTI	Enabled
MDS Mitigation	Inactive
Uptime	01 Hour 20 Minutes 35 Seconds
Current date/time	Mon Jun 22 22:28:06 IST 2026
DNS server(s)	127.0.0.1 ::1 8.8.8.8 1.1.1.1
Last config change	Mon Jun 22 21:40:56 IST 2026
State table size	0% (30/199000) Show states
MBUF Usage	0% (4572/1000000)
Load average	0.75, 0.60, 0.51
CPU usage	12%

Netgate Services And Support

Contract type: [Community Support](#)
[Community Support Only](#)

NETGATE AND pfSense COMMUNITY SUPPORT RESOURCES

If you purchased your pfSense gateway firewall appliance from Netgate and elected Community Support at the point of sale or installed pfSense on your own hardware, you have access to various community support resources. This includes the NETGATE RESOURCE LIBRARY.

You also may upgrade to a Netgate Global Technical Assistance Center (TAC) Support subscription. We're always on! Our team is staffed 24x7x365 and committed to delivering enterprise-class, worldwide support at a price point that is more than competitive when compared to others in our space.

- Upgrade Your Support
- Netgate Global Support FAQ
- Netgate Professional Services
- Community Support Resources
- Official pfSense Training by Netgate
- Visit Netgate.com

If you decide to purchase a Netgate Global TAC Support subscription, you **MUST** have your Netgate Device ID (NDI) from your firewall in order to validate support for this unit. Write down your NDI and store it in a safe place. You can purchase TAC support [here](#).

Interfaces

WAN	1000baseT- full-duplex	10.0.2.15 fd17:625c:f037:2:a00:27ff:fe3:a067
LAN	1000baseT- full-duplex	192.168.56.2

Install Squid and SquidGuard

← → ↻ Not secure https://192.168.56.2/pkg_mgr_install.php?pkg=pfSense-pkg-squid

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System Interfaces Firewall Services VPN Status Diagnostics Help

System / Package Manager / Package Installer

Installed Packages Available Packages **Package Installer**

Confirmation Required to install package pfSense-pkg-squid.

☒ Confirm

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← → ↻ Not secure https://192.168.56.2/pkg_mgr_install.php ☆ | 📌 | 📄 | ⌵ | 📄 All Bookmarks

pfSense COMMUNITY EDITION System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

System / Package Manager / Package Installer ?

pfSense-pkg-squid installation successfully completed.

Installed Packages Available Packages **Package Installer**

Package Installation Auto-scroll

```
configuration file /usr/local/etc/squid/squid.conf.sample.  
  
/usr/local/etc/squid/squid.conf.documented is a fully annotated  
configuration file you can consult for further reference.  
  
Additionally, you should check your configuration by calling  
'squid -f /path/to/squid.conf -k parse' before starting Squid.  
=====  
Message from pfSense-pkg-squid-0.5.3:  
  
--  
Please visit Services - Squid Proxy Server menu to configure the package and enable the proxy.  
>>> Cleaning up cache...done.  
Success
```

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← → ↻ Not secure https://192.168.56.2/pkg_mgr_install.php ☆ | 📌 | 📄 | ⌵ | 📄 All Bookmarks

pfSense COMMUNITY EDITION System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

System / Package Manager / Package Installer ?

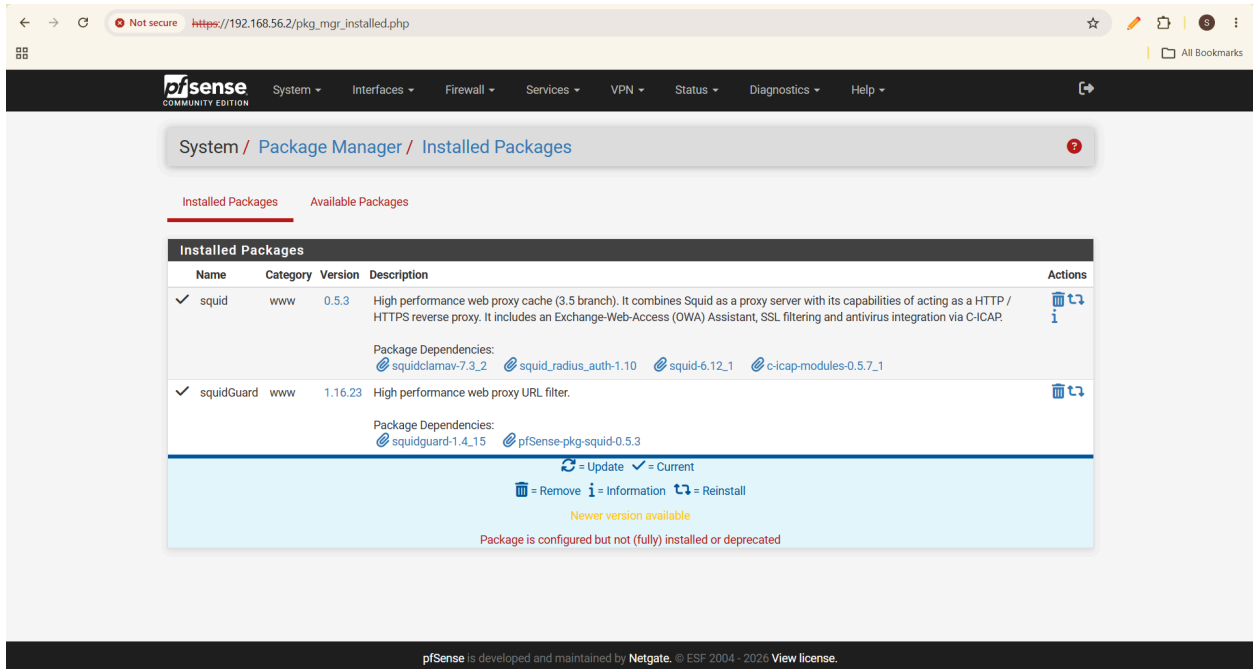
pfSense-pkg-squidGuard installation successfully completed.

Installed Packages Available Packages **Package Installer**

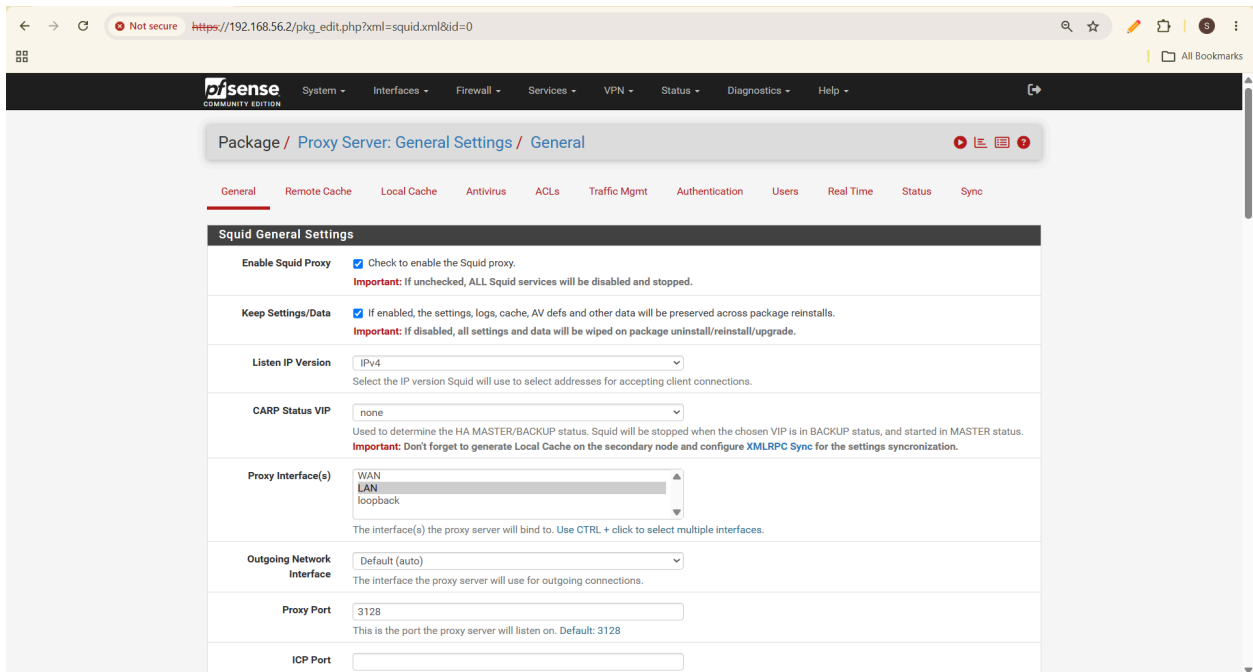
Package Installation Auto-scroll

```
Check documentation here:  
  
http://www.squidguard.org/Doc/  
  
To activate the changes do a /usr/local/sbin/squid -k reconfigure  
=====  
Message from pfSense-pkg-squidguard-1.16.23:  
  
--  
Please visit Services - SquidGuard Proxy Filter - Target Categories and set up at least one category there before enabling SquidGuard. See  
https://docs.netgate.com/pfsense/en/latest/packages/cache-proxy/squidguard.html for details.  
>>> Cleaning up cache...done.  
Success
```

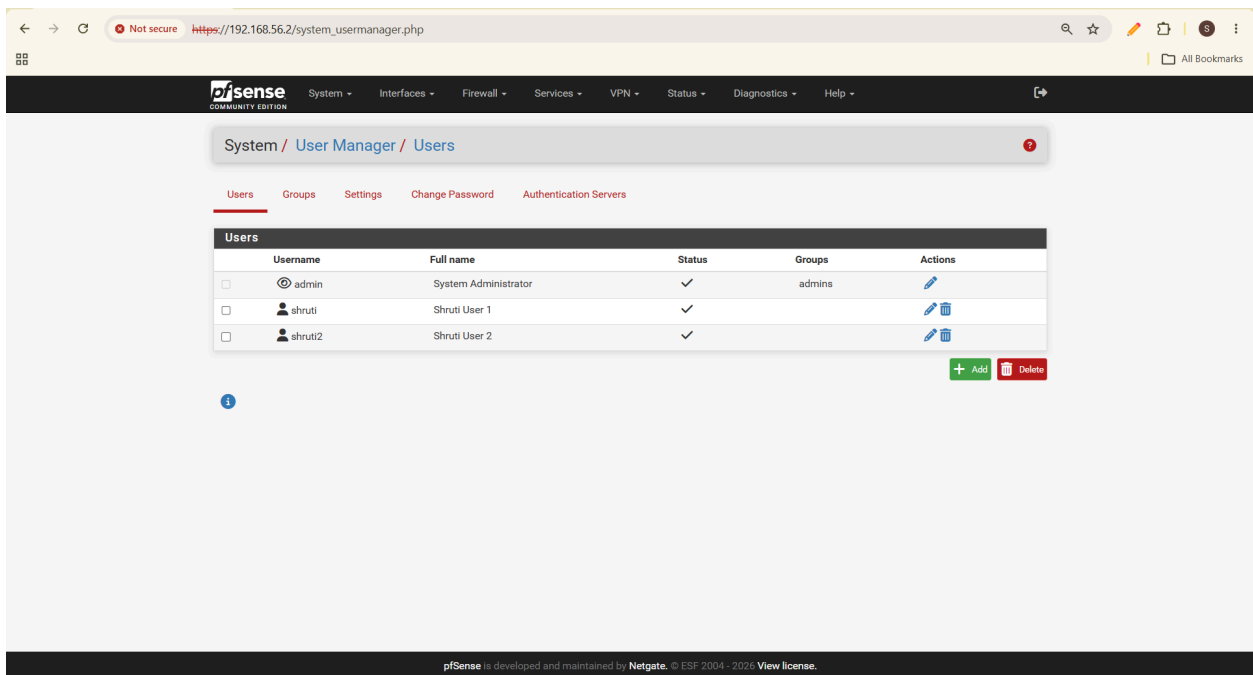
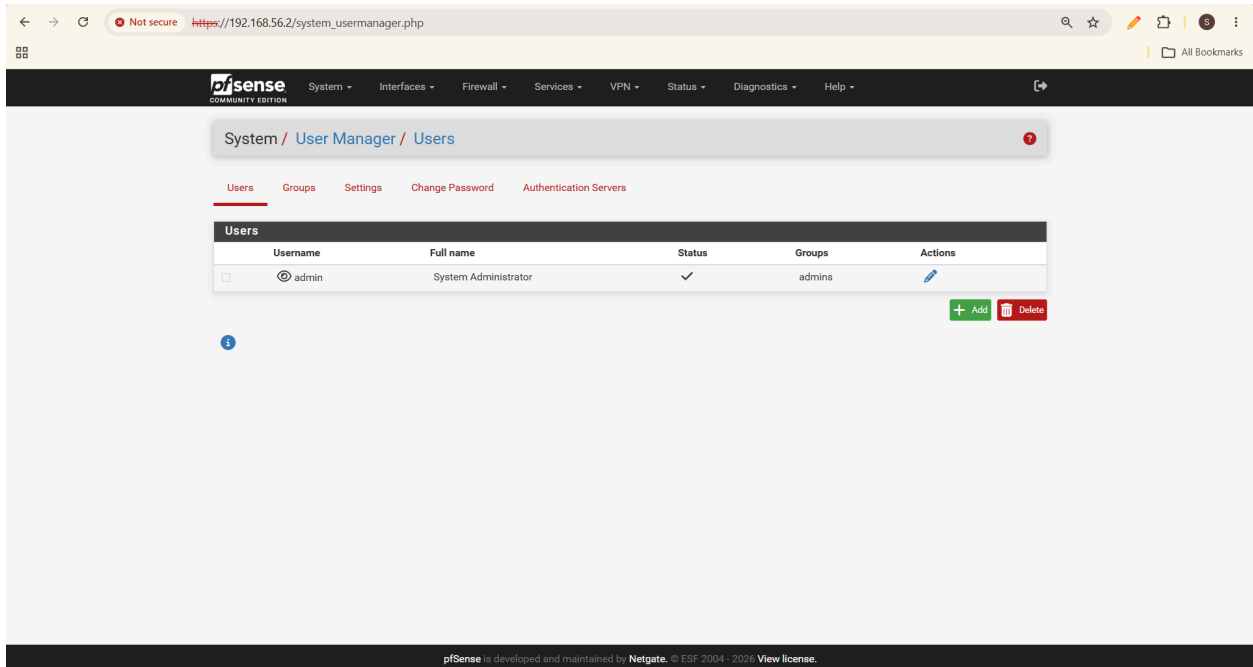
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Enable Squid Proxy



Create User Account



Configure Squid Authentication


System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

Package / Proxy Server: Authentication / Authentication

General Remote Cache Local Cache Antivirus ACLs Traffic Mgmt **Authentication** Users Real Time Status Sync

Squid Authentication General Settings

Authentication Method Local ▾
Select an authentication method. This will allow users to be authenticated by local or external services.

Authentication Server
Enter the IP or hostname of the server that will perform the authentication here.

Authentication server port
Enter the port to use to connect to the authentication server here.
Leave this field blank to use the authentication method's default port.

Authentication Prompt
This string will be displayed at the top of the authentication request window.

Authentication Processes
The number of authenticator processes to spawn. If many authentications are expected within a short timeframe, increase this number accordingly.

Authentication TTL
This specifies for how long (in minutes) the proxy server assumes an externally validated username and password combination to be valid.
When the Time To Live expires, the user will be prompted for credentials again. **Default: 5**

Authentication Max User IP Addresses
Enforces a limit to the number of unique IP addresses from which a single user can login.
Attempts to login from additional IP addresses are denied until the Authentication TTL has expired
Default: none ⓘ

Require Authentication for Unrestricted IPs ☐ If enabled, even 'Unrestricted IPs' configured on the ACLs tab are required to authenticate to use the proxy.

Subnets That Don't Need Authentication

Enable SquidGuard

← → 🔍 Not secure https://192.168.56.2/pkg_edit.php?xml=squidguard.xml&id=0


System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

Package / Proxy filter SquidGuard: General settings / General settings

General settings **Common ACL** Groups ACL Target categories Times Rewrites Blacklist Log XMLRPC Sync

General Options

Enable ☒ Check this option to enable squidGuard.
Important: Please set up at least one category on the 'Target Categories' tab before enabling. See [this link for details](#).
The Save button at the bottom of this page must be clicked to save configuration changes.
To activate squidGuard configuration changes, **the Apply button must be clicked.**

SquidGuard service state: **STARTED**

LDAP Options

Enable LDAP Filter ☐ Enable options for setup ldap connection to create filters with ldap search

LDAP DN
Configure your LDAP DN (ex: cn=Administrator,cn=Users,dc=domain)

LDAP DN Password
Password must be initialize with letters (Ex: Change123), valid format: [a-zA-Z][a-zA-Z0-9/_!\-.\%\{\}\+\?=&]

LDAP Cache Time
Number of seconds to cache LDAP Results (recommended value: 300)

Strip NT domain name ☐ Strip NT domain name component from user names (/ or \ separated).

Strip Kerberos Realm ☐ Strip Kerberos Realm component from user names (@ separated).

LDAP Version

Create Group ACL

← → ↻ Not secure https://192.168.56.2/pkg_edit.php?xml=squidguard_acl.xml&id=2

System - Interfaces - Firewall - Services - VPN - Status - Diagnostics - Help

Proxy filter SquidGuard: Groups Access Control List (ACL) / Edit / Groups ACL

General settings Common ACL **Groups ACL** Target categories Times Rewrites Blacklist Log XMLRPC Sync

General Options

Disabled ☐ Check this to disable this ACL rule.

Name

Enter a unique name of this rule here.
The name must consist between 2 and 15 symbols [a-z,0-9]. The first one must be a letter.

Order

Select the new position for this ACL item. ACLs are evaluated on a first-match source basis.
Note: Search for a suitable ACL by field 'source' will occur before the first match. If you want to define an exception for some sources (IP) from the IP range, put them on first of the list.
Example: ACL with single (or short range) source ip 10.0.0.15 must be placed before ACL with more large ip range 10.0.0.0/24.

Client (source)

Enter client's IP address or domain or "username" here. To separate them use space.
Example:
IP: 192.168.0.1 - Subnet: 192.168.0.0/24 or 192.168.1.0/255.255.255.0 - IP-Range: 192.168.1.1-192.168.1.10
Domain: foo.bar matches foo.bar or *.foo.bar
Username: 'user1'
Ldap search (Ldap filter must be enabled in General Settings):
ldapusersearch ldap:/192.168.0.190/DC=domain,DC=com?sAMAccountName?sub?(sAMAccountName=*)
(memberOf=CN=V%2cCN=Users%2cDC=domain%2cDC=com))
Attention: these line don't have break line, all on one line

Time

Select the time in which 'Target Rules' will operate or leave 'none' for rules without time restriction. If this option is set then in off-time the second ruleset will operate.

Target Rules

← → ↻ Not secure https://192.168.56.2/pkg_edit.php?xml=squidguard_acl.xml&id=2

System - Interfaces - Firewall - Services - VPN - Status - Diagnostics - Help

Proxy filter SquidGuard: Groups Access Control List (ACL) / Edit / Groups ACL

General settings Common ACL **Groups ACL** Target categories Times Rewrites Blacklist Log XMLRPC Sync

General Options

Disabled ☐ Check this to disable this ACL rule.

Name

Enter a unique name of this rule here.
The name must consist between 2 and 15 symbols [a-z,0-9]. The first one must be a letter.

Order

Select the new position for this ACL item. ACLs are evaluated on a first-match source basis.
Note: Search for a suitable ACL by field 'source' will occur before the first match. If you want to define an exception for some sources (IP) from the IP range, put them on first of the list.
Example: ACL with single (or short range) source ip 10.0.0.15 must be placed before ACL with more large ip range 10.0.0.0/24.

Client (source)

Enter client's IP address or domain or "username" here. To separate them use space.
Example:
IP: 192.168.0.1 - Subnet: 192.168.0.0/24 or 192.168.1.0/255.255.255.0 - IP-Range: 192.168.1.1-192.168.1.10
Domain: foo.bar matches foo.bar or *.foo.bar
Username: 'user1'
Ldap search (Ldap filter must be enabled in General Settings):
ldapusersearch ldap:/192.168.0.190/DC=domain,DC=com?sAMAccountName?sub?(sAMAccountName=*)
(memberOf=CN=V%2cCN=Users%2cDC=domain%2cDC=com))
Attention: these line don't have break line, all on one line

Time

Select the time in which 'Target Rules' will operate or leave 'none' for rules without time restriction. If this option is set then in off-time the second ruleset will operate.

Target Rules

Apply SquidGuard Configuration

The screenshot shows the pfSense web interface for the SquidGuard configuration. The browser address bar indicates the URL `https://192.168.56.2/pkg_edit.php?xml=squidguard.xml&id=0`. The page title is "Package / Proxy filter SquidGuard: General settings / General settings". The "General settings" tab is selected, showing the "General Options" section. The "Enable" checkbox is checked, and a green "Apply" button is visible. Below the "Enable" section, the "SquidGuard service state" is shown as "STARTED". The "LDAP Options" section is also visible, with fields for "Enable LDAP Filter", "LDAP DN", "LDAP DN Password", "LDAP Cache Time", "Strip NT domain name", and "Strip Kerberos Realm".

Package / Proxy filter SquidGuard: General settings / General settings

General settings Common ACL Groups ACL Target categories Times Rewrites Blacklist Log XMLRPC Sync

General Options

Enable ☒ Check this option to enable squidGuard.
Important: Please set up at least one category on the 'Target Categories' tab before enabling. See [this link](#) for details.
The Save button at the bottom of this page must be clicked to save configuration changes.
To activate squidGuard configuration changes, **the Apply button must be clicked.**
[Apply](#)

SquidGuard service state: **STARTED**

LDAP Options

Enable LDAP Filter ☐ Enable options for setup ldap connection to create filters with ldap search

LDAP DN
Configure your LDAP DN (ex: cn=Administrator,cn=Users,dc=domain)

LDAP DN Password
Password must be initialize with letters (Ex: Change123), valid format: [a-zA-ZV/[a-zA-Z0-9/_!~\%!*+?=&]

LDAP Cache Time
Number of seconds to cache LDAP Results (recommended value: 300)

Strip NT domain name ☐ Strip NT domain name component from user names (/ or \ separated).

Strip Kerberos Realm ☐ Strip Kerberos Realm component from user names (@ separated).

Create Target Category for Social, shruti1, shruti2 websites

The screenshot shows the pfSense web interface for the SquidGuard configuration. The browser address bar indicates the URL `https://192.168.56.2/pkg.php?xml=squidguard_dest.xml`. The page title is "Package / Proxy filter SquidGuard: Target categories / Target categories". The "Target categories" tab is selected, showing a table of target categories. The table has columns for "Name", "Redirect", and "Description". The categories listed are "blocktest", "social", "shruti2sites", "shruti1sites", and "commonsites". Each category has edit and delete icons. A green "Add" button is at the bottom right of the table. A "Save" button is at the bottom left of the page.

Package / Proxy filter SquidGuard: Target categories / Target categories

General settings Common ACL Groups ACL Target categories Times Rewrites Blacklist Log XMLRPC Sync

Name	Redirect	Description
blocktest		
social		
shruti2sites		
shruti1sites		
commonsites		

[Save](#) [Add](#)

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← → ↻ Not secure https://192.168.56.2/pkg_edit.php?xml=squidguard_default.xml&id=0 ☆ ⚙️ 📌 All Bookmarks

Package / Proxy filter SquidGuard: Common Access Control List (ACL) / Common ACL ?

General settings **Common ACL** Groups ACL Target categories Times Rewrites Blacklist Log XMLRPC Sync

General Options

Target Rules |blocktest shruti2sites shruti1sites commonsites fall

Target Rules List + -

ACCESS: 'whitelist' - always pass; 'deny' - block; 'allow' - pass, if not blocked.

Target Categories

[blocktest]	access	deny	▼
[social]	access	---	▼
[shruti2sites]	access	allow	▼
[shruti1sites]	access	allow	▼
[commonsites]	access	allow	▼
Default access [all]	access	deny	▼

Do not allow IP-Addresses in URL ☐ To make sure that people do not bypass the URL filter by simply using the IP-Addresses instead of the FQDN you can check this option. This option has no effect on the whitelist.

Proxy Denied Error

The first part of the error message displayed to clients when access was denied. Defaults to **Request denied by g_get(product_name) proxy.**

Different mode

Configure Access Policy

← → ↻ Not secure https://192.168.56.2/pkg_edit.php?xml=squidguard_acl.xml&act=edit&id=1 🔍 ☆ ⚙️ 📌 All Bookmarks

Name |shruti1|

Enter a unique name for this rule here.
The name must consist between 2 and 15 symbols [a-Z,0-9]. The first one must be a letter.

Order |shruti2|

Select the new position for this ACL item. ACLs are evaluated on a first-match source basis.
Note:
Search for a suitable ACL by field 'source' will occur before the first match. If you want to define an exception for some sources (IP) from the IP range, put them on first of the list.
Example:
ACL with single (or short range) source ip 10.0.0.15 must be placed before ACL with more large ip range 10.0.0.0/24.

Client (source) |shruti1|

Enter client's IP address or domain or "username" here. To separate them use space.
Example:
IP: 192.168.0.1 - Subnet: 192.168.0.0/24 or 192.168.1.0/255.255.255.0 - IP-Range: 192.168.1.1-192.168.1.10
Domain: foo.bar matches foo.bar or *.foo.bar
Username: user1
Ldap search (Ldap filter must be enabled in General Settings):
ldapsu=search ldap://192.168.0.100/DC=domain,DC=com?sAMAccountName?sub?(&(sAMAccountName=%s)(memberOf=CN=H%2cCN=Users%2cDC=domain%2cDC=com))
Attention: these line don't have break line, all on one line

Time |socialtime|

Select the time in which 'Target Rules' will operate or leave 'none' for rules without time restriction. If this option is set then in off-time the second ruleset will operate.

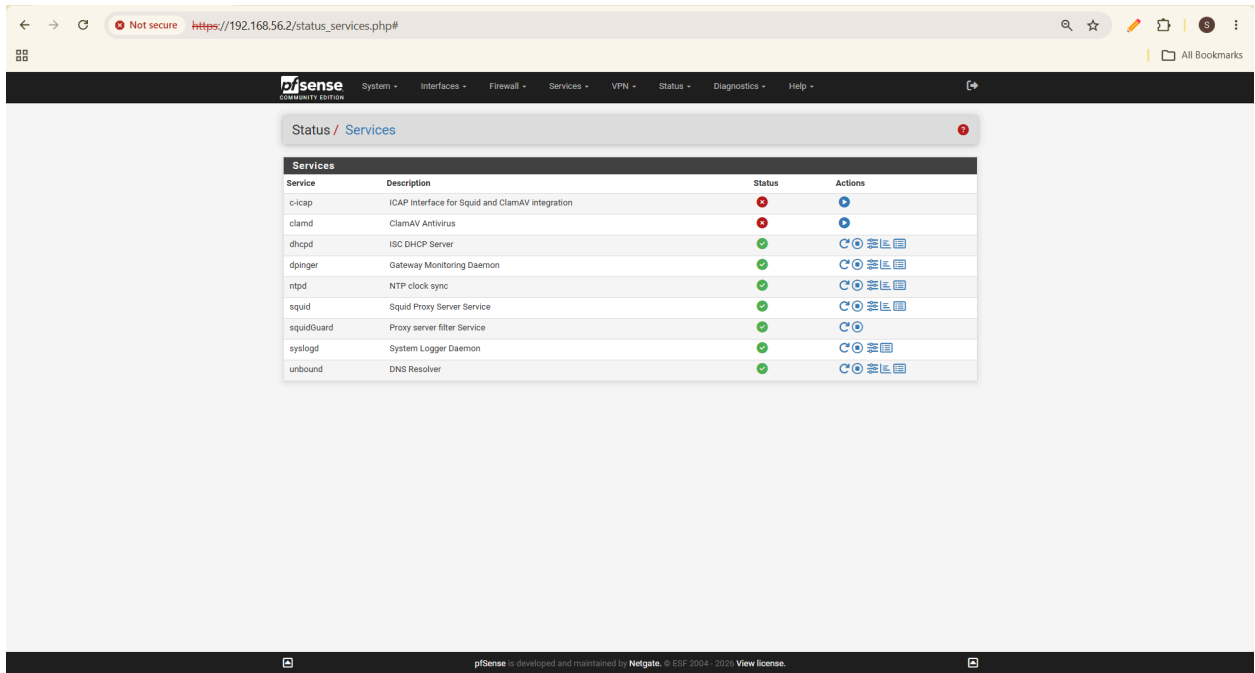
Target Rules |blocktest all [|blocktest all|

Target Rules List + -

ACCESS: 'whitelist' - always pass; 'deny' - block; 'allow' - pass, if not blocked.

Target Categories				Target Categories for off-time			
				if Time not defined, this is column will be ignored			
[blocktest]	access	deny	▼	[blocktest]	access	deny	▼
[social]	access	allow	▼	[social]	access	deny	▼
[shruti2sites]	access	deny	▼	[shruti2sites]	access	---	▼
[shruti1sites]	access	allow	▼	[shruti1sites]	access	---	▼
[commonsites]	access	allow	▼	[commonsites]	access	---	▼
Default access [all]	access	allow	▼	Default access [all]	access	allow	▼

Restart the services in Status tab



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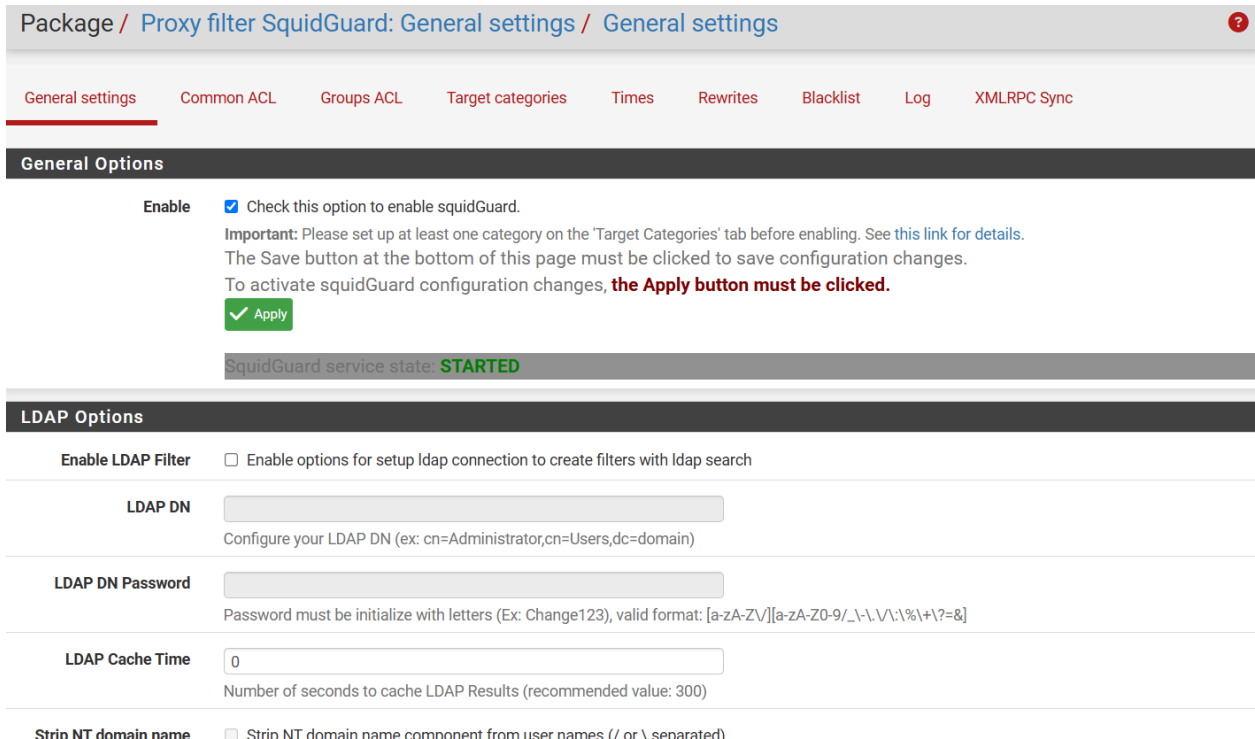
System • Interfaces • Firewall • Services • VPN • Status • Diagnostics • Help

Status / Services

Service	Description	Status	Actions
c-icap	ICAP Interface for Squid and ClamAV integration	✗	↻
clamd	ClamAV Antivirus	✗	↻
dhcpd	ISC DHCP Server	✓	↻ ⏏ ⏏ ⏏
dpinger	Gateway Monitoring Daemon	✓	↻ ⏏ ⏏ ⏏
ntpd	NTP clock sync	✓	↻ ⏏ ⏏ ⏏
squid	Squid Proxy Server Service	✓	↻ ⏏ ⏏ ⏏
squidGuard	Proxy server filter Service	✓	↻ ⏏
syslogd	System Logger Daemon	✓	↻ ⏏ ⏏
unbound	DNS Resolver	✓	↻ ⏏ ⏏ ⏏

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Apply SquidGuard Configuration



Package / Proxy filter SquidGuard: General settings / General settings

General settings Common ACL Groups ACL Target categories Times Rewrites Blacklist Log XMLRPC Sync

General Options

Enable ☒ Check this option to enable squidGuard.
Important: Please set up at least one category on the 'Target Categories' tab before enabling. See [this link for details](#).
The Save button at the bottom of this page must be clicked to save configuration changes.
To activate squidGuard configuration changes, **the Apply button must be clicked.**

✓ Apply

SquidGuard service state: **STARTED**

LDAP Options

Enable LDAP Filter ☐ Enable options for setup ldap connection to create filters with ldap search

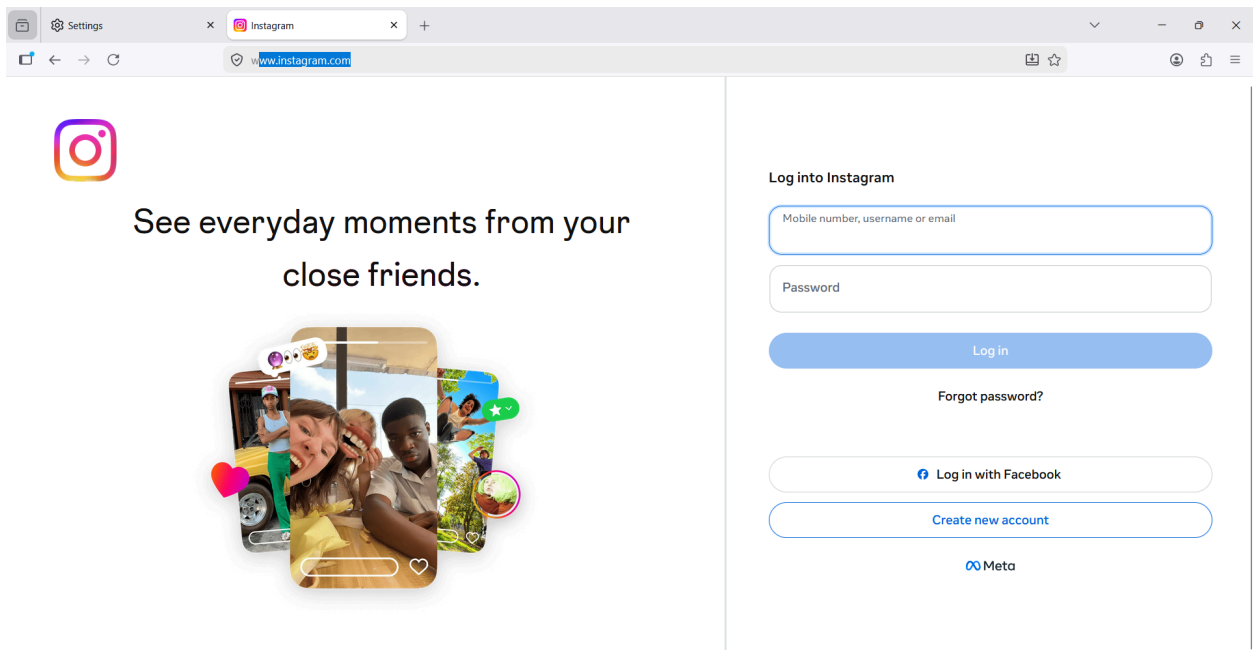
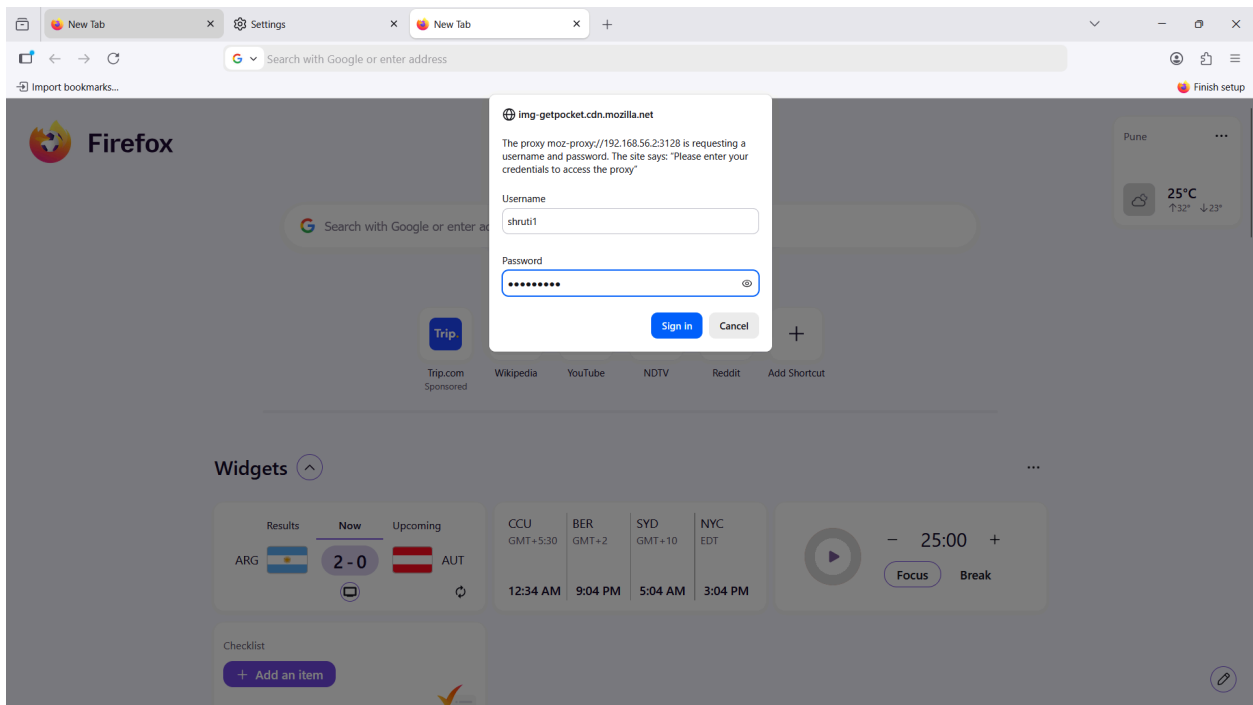
LDAP DN
Configure your LDAP DN (ex: cn=Administrator,cn=Users,dc=domain)

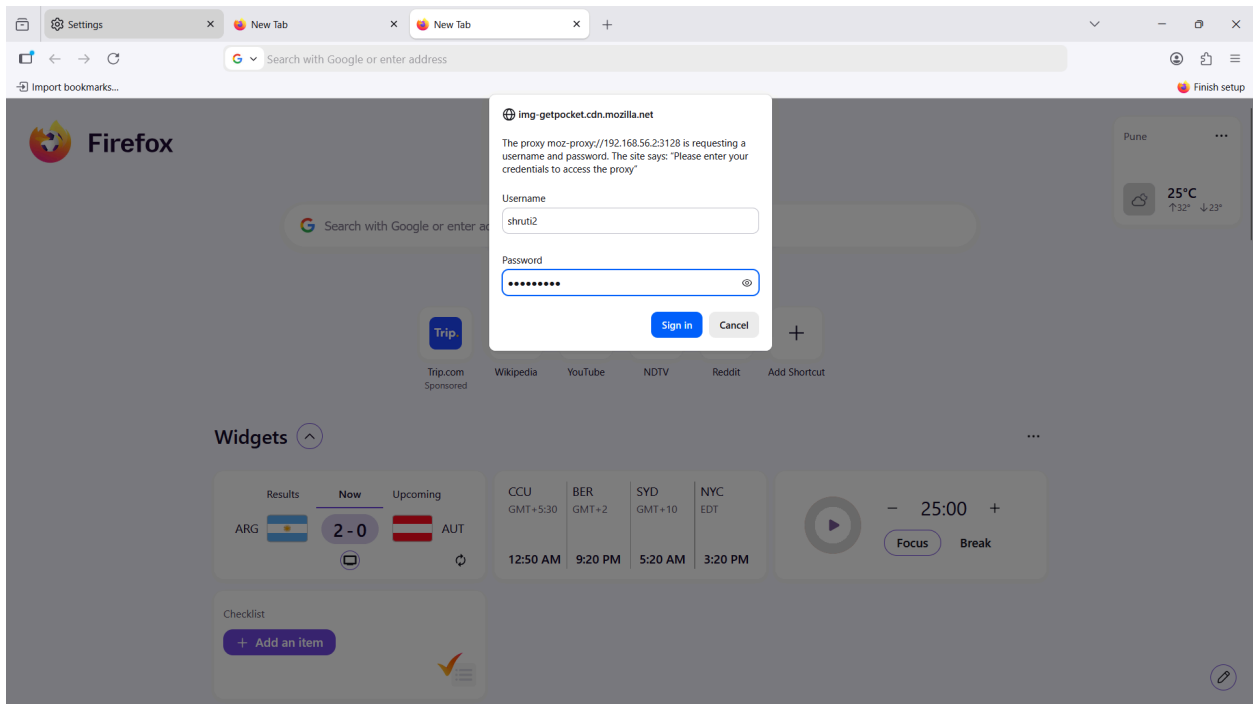
LDAP DN Password
Password must be initialize with letters (Ex: Change123), valid format: [a-zA-Z][a-zA-Z0-9/_!~\.\%\\+?=&]

LDAP Cache Time
Number of seconds to cache LDAP Results (recommended value: 300)

Strin NT domain name ☐ Strin NT domain name component from user names (/ or \ separated)

Test Access Using all Users





img-getpocket.cdn.mozilla.net

The proxy moz-proxy://192.168.56.2:3128 is requesting a username and password. The site says: "Please enter your credentials to access the proxy"

Username

shrutiz

Password

Sign In Cancel

Results	Now	Upcoming
ARG	2-0	AUT

CCU	BER	SYD	NYC
GMT+5:30	GMT+2	GMT+10	EDT
12:50 AM	9:20 PM	5:20 AM	3:20 PM

25:00

Focus Break

Checklist

+ Add an item

